Counterfactual Analysis for Adverse Drug Event Risk Prediction

Raksha Basnet, Max Thacker, Cole Johnson, Shahe Shahinian

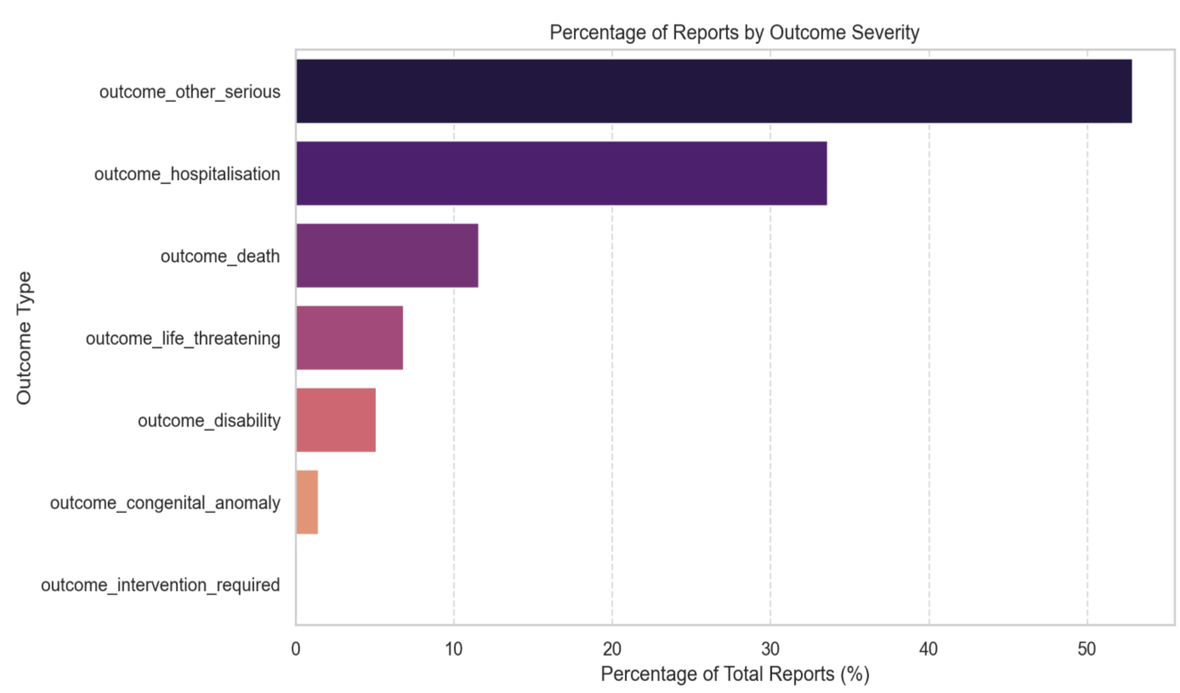
Summary

How do doctors choose the right drugs for their patient? *What* risks do those choices carry? *Do* some treatments cause hidden harm for certain patients while helping others? In the U.S. alone, adverse effects cause over **1.5 million emergency visits** annually, putting strain on the medical system and causing patient harm. In response, we combined visualization and machine learning to **empower clinicians to answer these questions**. We analyzed adverse events reported to the FDA from 2012–2025 and compared the treatments patients received with alternative treatments they might have been given.

Dataset

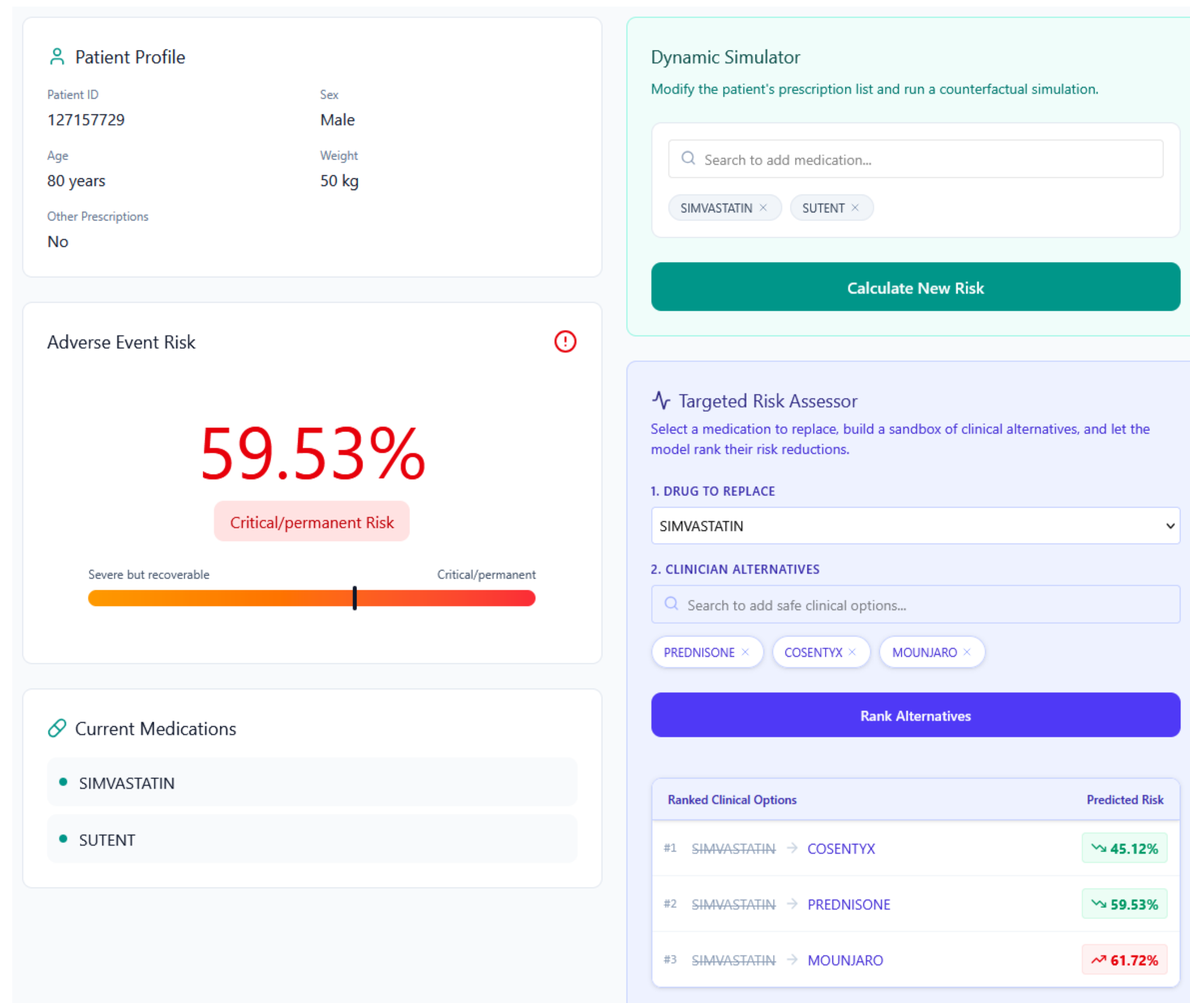
The U.S. Food and Drug Administration collects reports of suspected drug side effects in the **Adverse Event Reporting System**, a public safety database. We analyzed reports from Q4 2012 through Q4 2025 forming **tens of millions of reports, occupying 15GB of disk space**. Each report includes information such as drugs taken, patient characteristics, and reported adverse events.

Distribution of Outcome Severities



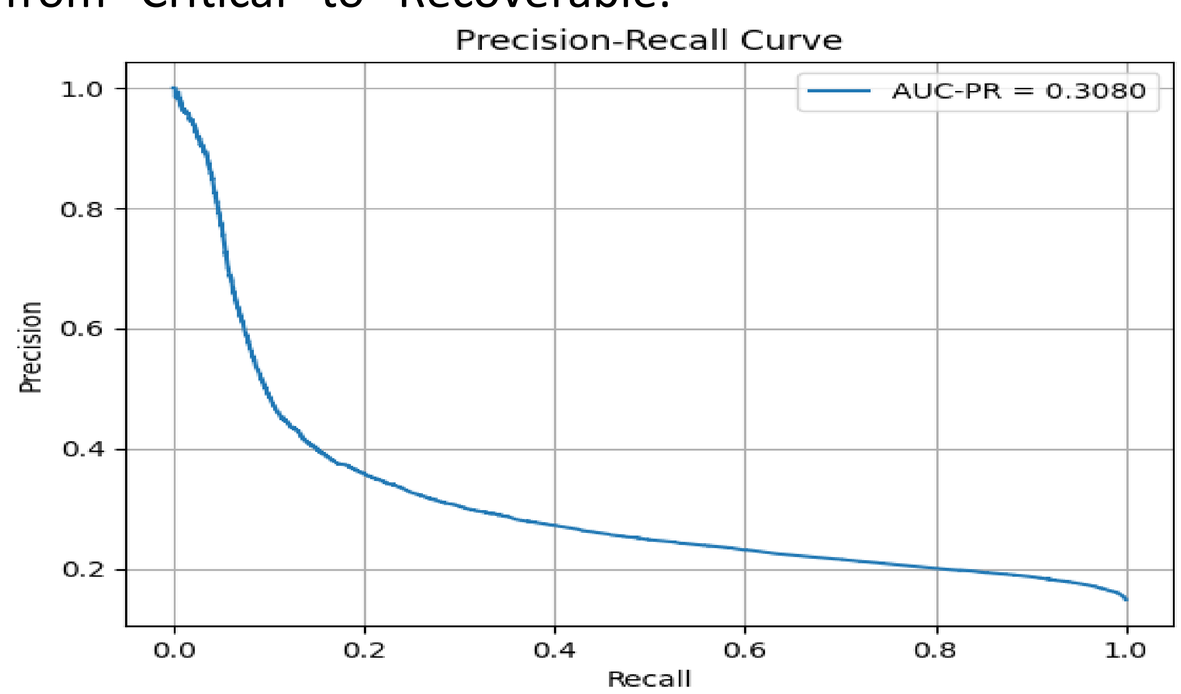
Approaches

Our approach integrates an **XGBoost classifier** with a post-hoc **counterfactual analysis** pipeline to predict and interpret adverse drug event risks. The model categorizes outcomes as "critical" or "recoverable," while the counterfactual component simulates "what-if" scenarios by modifying drug exposures to observe real-time risk shifts. By prioritizing **clinical recall over raw accuracy**, this strategy ensures rare, life-threatening signals are captured and clinically explained. This transition from a "black-box" model to an interactive tool allows clinicians to quantify medication influence, providing a **state-of-the-art solution for high-stakes medical monitoring**.



Counterfactual Risk Sandbox

The **MedRisk optimizer** clinician dashboard displays a patient's risk based on their demographic and drug information. The clinician can choose a drug to replace and see in real time a projected change to their risk of having an adverse reaction.

Experiments & Results

Our framework reliably distinguishes between patients likely to recover and those at high risk or permanent harm, maintaining a consistent sensitivity of around 0.70 across diverse patient populations. We compared our optimized model against an untuned baseline, using **Youden's J Statistic** to recalibrate the decision threshold to **0.49** for maximum clinical sensitivity. This resulted in a significant performance leap, achieving an **AUCPR of 0.32** and outperforming standard configurations that suffered from high false-negative rates due to class imbalance. Finally, **counterfactual analysis** validated the tool's utility for simulating how drug substitutions shift patient risk from "Critical" to "Recoverable."